

Flashing the Factory Firmware

1. Objectives of This Section

Use the flash_download_tool utility to flash the factory firmware (MicroROS-V2.bin) to the MicroROS-V2 control board (ESP32-S3).

2. Hardware Requirements

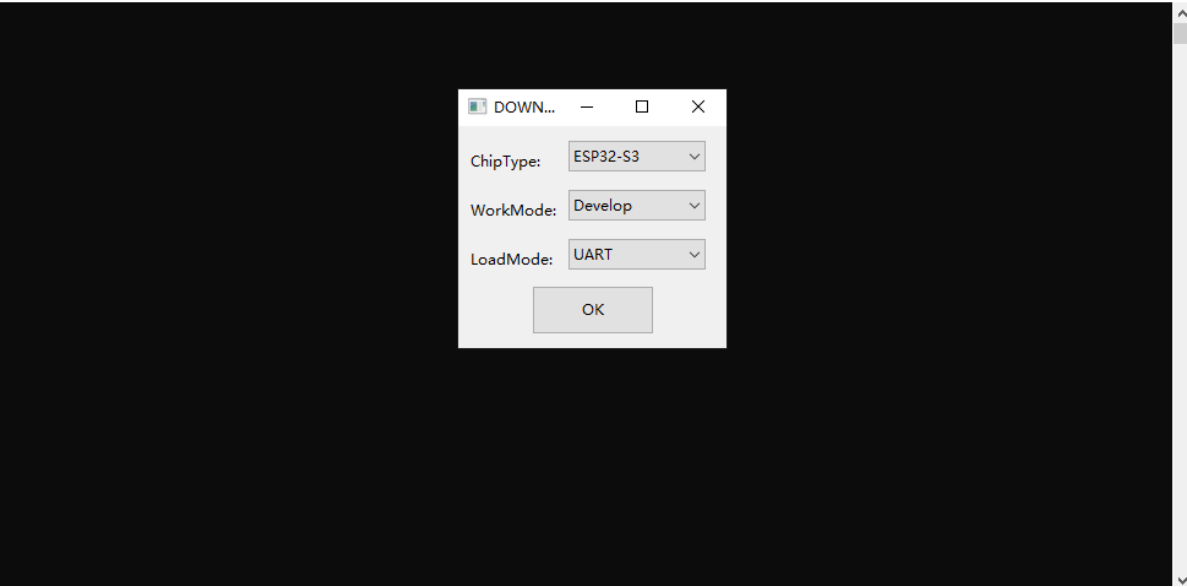
Hardware	Description
MicroROS-V2 control board	ESP32-S3 main controller
USB data cable	Connects the control board and the computer
Computer	With flash_download_tool installed

3. Flashing Steps

1. Select the Chip and Work Mode

Open flash_download_tool and make the following settings:

- **Chip model:** ESP32-S3
- **WorkMode:** Develop
- **LoadMode:** UART

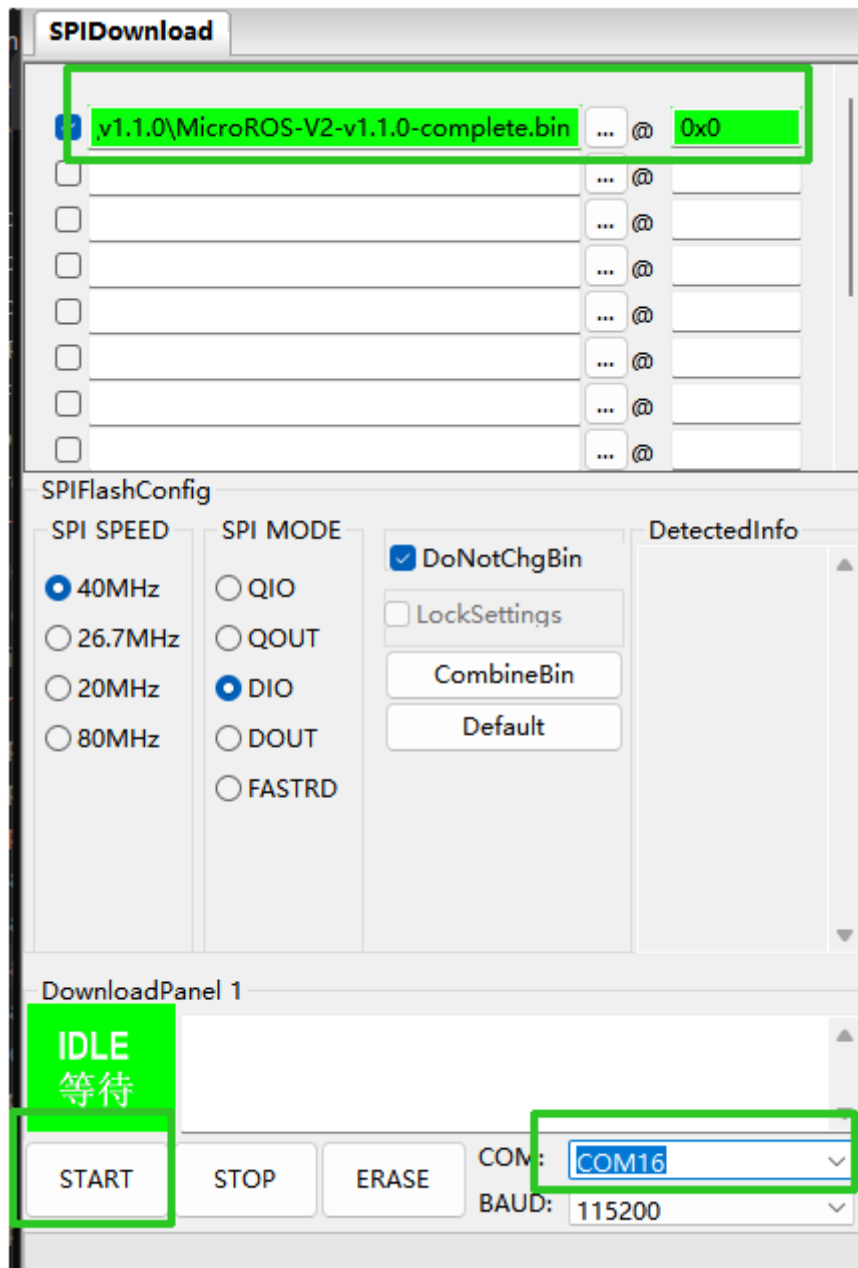


2. Configure the Flashing Parameters

Load the firmware and configure the SPI parameters:

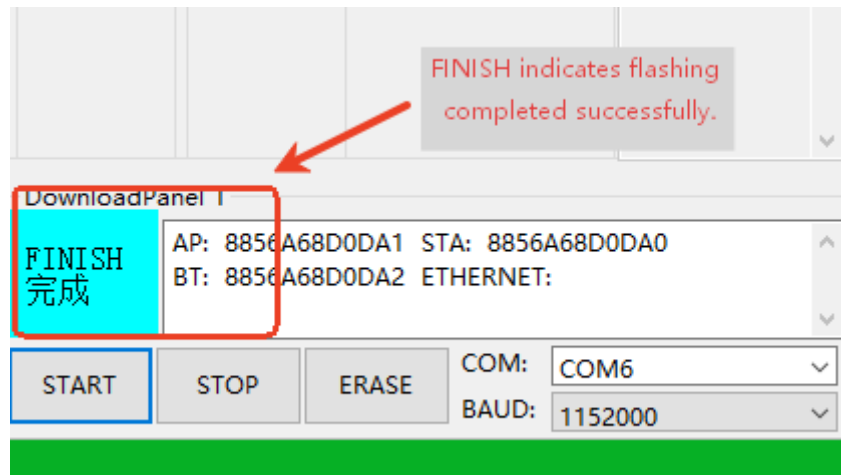
Parameter	Setting Value	Description
bin file	MicroROS-V2.bin	Factory test firmware; the firmware name may differ by version number, but the flashing procedure is the same
Start address	0x0	The start address for flashing the firmware
SPI mode	QIO	DIO can be selected in compatibility mode
SPI SPEED	40MHz	SPI communication speed
COM port	Corresponding serial port	Select according to the serial port actually connected

After configuration, click **START** to begin flashing.



3. Wait for the Flashing to Complete

During flashing, wait for the progress bar to finish; the **FINISH** prompt indicates that flashing succeeded. You can then reset the main control board and the firmware will take effect.



4. Precautions

- Before flashing, make sure the control board is connected to the computer via a USB cable and the corresponding serial port is recognized
- If flashing fails, check that the USB cable supports data transfer (not a charge-only cable)
- Prefer the QIO SPI mode; switch to DIO if you encounter compatibility issues